Goldi

+91-8527302147 | grimissher@gmail.com | New Delhi, India
GitHub | LinkedIn | LeetCode | Portfolio

Education

Delhi Technological University (DTU) | *B.Tech in Computer Science* | **CGPA: 7.65** (Roll: 23/CS/158) 2023 – 2027

School Of Excellence, Khichripur, Delhi | *CBSE Class XII:* ***93.4%*** | *CBSE Class X:* ***91.6%*** 2021 – 2023

Work Experience

AI Coding & Model Evaluator (Freelance) | *Outlier AI (Scale AI)* June 2026 – Present

- Evaluated **AI-generated code completions** across Python and C++ for algorithmic correctness, time/space complexity, and edge-case resilience.
- Conducted rubric-based **RLHF evaluations**, benchmarking multi-model outputs against strict criteria for instruction compliance, logic soundness, and hallucination reduction.
- Authored detailed technical rationales and designed **adversarial test inputs** to isolate failure modes, optimize reasoning chains, and assist LLM fine-tuning pipelines.

Projects

AegisGateway – Resilient Distributed LLM Gateway | *FastAPI, Redis, Docker, LangChain, BGE* GitHub

- Engineered a distributed LLM reverse proxy in FastAPI and Redis, implementing a sliding-window log rate limiter (ZSET) to prevent boundary burst spikes with ∼1ms Redis pipeline execution overhead.
- Architected a two-stage Semantic Caching engine using BGE-small embeddings and an $O(1)$ lexical polarity guard, reducing **cache-hit p95 latency from 2140ms to 14ms (99.3% drop)** while eliminating negation false-positives.
- Designed a fault-tolerant fallback router with a distributed Circuit Breaker pattern in Redis, enabling automated zero-downtime failover from degraded local endpoints (Llama 3.1) to backup cloud endpoints (Gemma 2).

SpecuRAG – Speculative RAG Engine | *Python, Ollama, ChromaDB, BM25, Scikit-learn, AsyncIO* GitHub

- **Speculative RAG Pipeline**: Built an asynchronous Speculative RAG system based on ICLR 2025 research, decoupling contextual drafting (Llama-3.2-3B) from verification (Llama-3.1-8B open-weights model).
- **Hybrid Retrieval & Rank Fusion**: Engineered a dual-channel retrieval engine combining BM25 keyword search with dense embeddings (nomic-embed-text) fused via Reciprocal Rank Fusion (RRF, $k = 60$).
- **Dynamic Semantic Partitioning**: Implemented K-Means clustering with unsupervised Silhouette-score k-selection ($k \in \{2, 3\}$) and inverted modulo snake allocation to balance evidence quality across drafters.
- **Prompt Compression & Bias Mitigation**: Mitigated long-context "lost-in-the-middle" position bias and reduced verifier prompt size by ∼85% by evaluating answer-rationale pairs (α, β) without re-reading raw chunks.

AutoData – Adversarial QA Dataset Generator | *LangGraph, DeepEval, Ollama, ChromaDB, LangSmith* GitHub

- Architected an adversarial multi-agent pipeline inspired by Meta FAIR's "Autodata" research to autonomously generate and validate multi-hop reasoning QA datasets for model fine-tuning.
- Implemented a Weak-Strong evaluation loop (Llama 3.1 8B vs. Gemma 2 27B, exhibiting a **20+ point benchmark accuracy gap**), Corrective RAG (CRAG), and **DeepEval** metrics for Faithfulness, Precision, and Relevancy.
- Designed iterative difficulty scaling across up to 5 refinement loops with end-to-end LangSmith distributed tracing across a Challenger-Solver-Judge architecture.

Open-Source & Systems

`pip install decmas` | *Python, LangChain, Pydantic, Concurrency* PyPI | GitHub

- Architected and published an open-source framework inspired by Stanford's DeLM research, resolving "Router-Agent bottlenecks" via a decentralized peer-to-peer agent topology.
- Engineered an Asynchronous Task Queue with concurrency controls and an LLM-as-a-Verifier layer, eliminating cascading hallucinations and **reducing redundant API calls by 50%**.
- Published package to PyPI with modular abstractions, Pydantic type validation, and comprehensive unit test suites for multi-agent workflows.

Technical Skills

Languages: Python, SQL, C++
GenAI & Evaluation: LangGraph, LangChain, DeepEval, G-Eval, RAG / Corrective RAG, Semantic Caching, ChromaDB, FAISS, Ollama
Backend & Infrastructure: FastAPI, Redis (ZSET, Hashes), PostgreSQL, Docker, REST APIs, Microservices Architecture
Developer Tools & Libraries: Git, GitHub, LangSmith, Pydantic, Streamlit, Supabase
Core CS Fundamentals: Database Management Systems, Operating Systems, Computer Networks, Object-Oriented Programming

Achievements & Leadership

Competitive Programming: Knight badge on LeetCode (Rating 1900+, Top 3%). Solved **1200+ problems** across LeetCode, CSES, and GeeksforGeeks using C++.

Hackathons: Finalist at HACKJMI (Feb 2025) – Engineered an accessible mental health assessment tool using full-stack web technologies.

Campus Leadership: Active member of DTU Art & Poetry Society; In-Game Leader (IGL) & Champion at Inter-College Esports Tournament.